Supplementary Material

Q-LCAN: Anatomy-Constrained Quantum-Agentic Systems for MRI-Derived Structural Surgical Digital Twins in Uterine Adenomyosis

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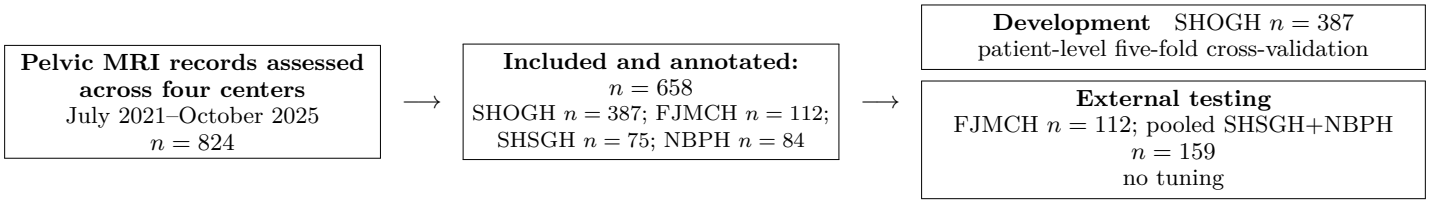
Supplementary Fig. S1. Patient selection and analysis allocation



Excluded ($n = 166$): incomplete/corrupted imaging or incomplete uterine coverage ($n = 43$); severe motion or metal artifact ($n = 28$); slice thickness > 5 mm ($n = 22$); missing pathology or baseline record ($n = 31$); major distortion from prior surgery or congenital malformation ($n = 18$); malignancy or severe systemic disease ($n = 24$).

Figure S1: Screening, exclusion, inclusion, and patient-level analysis allocation. Exclusion categories are mutually exclusive.

Supplementary Table S1. Clinical and pathological characteristics

Characteristic	SHOGH ($n = 387$)	FJMCH ($n = 112$)	SHSGH ($n = 75$)	NBPH ($n = 84$)	Total ($n = 658$)
Pathological confirmation					
Hysterectomy specimen	285 (73.6)	93 (83.0)	62 (82.7)	71 (84.5)	511 (77.7)
Adenomyotic-lesion excision specimen	102 (26.4)	19 (17.0)	13 (17.3)	13 (15.5)	147 (22.3)
MRI-to-surgery interval, median (IQR), days	18 (10–31)	21 (12–35)	16 (9–29)	19 (11–33)	18 (10–32)
Imaging phenotype					
Focal adenomyosis	112 (28.9)	38 (33.9)	22 (29.3)	28 (33.3)	200 (30.4)
Diffuse adenomyosis	184 (47.5)	49 (43.8)	36 (48.0)	37 (44.0)	306 (46.5)
Mixed adenomyosis	91 (23.5)	25 (22.3)	17 (22.7)	19 (22.6)	152 (23.1)
Symptoms and reproductive history					
Dysmenorrhea	315 (81.4)	91 (81.3)	61 (81.3)	68 (81.0)	535 (81.3)
Heavy menstrual bleeding	233 (60.2)	72 (64.3)	43 (57.3)	52 (61.9)	400 (60.8)
Chronic pelvic pain	118 (30.5)	39 (34.8)	25 (33.3)	29 (34.5)	211 (32.1)
Infertility/subfertility history	108 (27.9)	29 (25.9)	18 (24.0)	22 (26.2)	177 (26.9)
Nulliparous	143 (37.0)	43 (38.4)	25 (33.3)	29 (34.5)	240 (36.5)
Current fertility intention	174 (45.0)	54 (48.2)	31 (41.3)	37 (44.0)	296 (45.0)
Previous adenomyosis-directed treatment					
Hormonal treatment	171 (44.2)	52 (46.4)	31 (41.3)	36 (42.9)	290 (44.1)
Ablation	26 (6.7)	8 (7.1)	5 (6.7)	7 (8.3)	46 (7.0)
Uterus-preserving surgery	35 (9.0)	9 (8.0)	7 (9.3)	8 (9.5)	59 (9.0)
Coexisting benign pelvic disease					
Uterine fibroids	151 (39.0)	45 (40.2)	28 (37.3)	34 (40.5)	258 (39.2)
Endometriosis	66 (17.1)	20 (17.9)	12 (16.0)	13 (15.5)	111 (16.9)
Other benign pelvic disease	37 (9.6)	10 (8.9)	7 (9.3)	8 (9.5)	62 (9.4)

Values are n (%) unless otherwise stated. Symptoms, previous treatments, and comorbidities are not mutually exclusive. Phenotype was assigned by consensus review of the complete diagnostic MRI examination.

Supplementary Table S2. Center-specific MRI acquisition and examination variables

Acquisition variable	SHOGH ($n = 387$)	FJMCH ($n = 112$)	SHSGH ($n = 75$)	NBPH ($n = 84$)
Vendor	Siemens	GE	Philips	Siemens
Axial T2	TSE	FRFSE	TSE	TSE
Field	3 T	3 T	1.5 T	3 T
TR, ms	4500 (3800–6200)	5200 (4200–6800)	4800 (3600–6500)	4600 (3800–6000)
TE, ms	96 (86–112)	105 (92–120)	100 (90–120)	98 (88–110)
ETL	24 (20–32)	22 (18–28)	20 (16–28)	24 (20–30)
FOV, mm	240 (220–280)	250 (240–300)	250 (240–280)	260 (240–300)
Matrix; slice/gap, mm	320 × 320; 4.0/0.4	320 × 288; 4.0/0.4	352 × 320; 3.5/0.35	360 × 320; 3.5/0.35

TR, TE, ETL, and FOV are reported as median (range). TSE, turbo spin echo; FRFSE, fast-recovery fast spin echo; ETL, echo-train length; FOV, field of view.

Status at MRI	SHOGH	FJMCH	SHSGH	NBPH	Total
Menstruation	45	14	8	11	78 (11.9)
Follicular phase	146	42	28	33	249 (37.8)
Luteal phase	161	45	31	31	268 (40.7)
Amenorrhea/indeterminate phase	12	2	3	3	20 (3.0)
Postmenopausal	23	9	5	6	43 (6.5)
Active hormonal treatment	64	20	12	15	111 (16.9)

Cycle categories are mutually exclusive and sum to 658. Active hormonal treatment denotes treatment at the time of MRI and is reported separately from previous hormonal treatment in Table S1.